AN OBSTRUCTION THEORY WITH COUNTABLE COMPATIBILITY GIVES OPENNESS OF VERSALITY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3948-3975 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3977-4028, the complete author proof (50 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_artin.json (pb=2196, stmt 3948-3975, proof 3977-4028).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07YG: sources/stacks-artin/pinned-original-source.tex, source lines 3886–3905. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CXU: sources/stacks-artin/pinned-original-source.tex, source lines 3282–3293. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07Y9: sources/stacks-artin/pinned-original-source.tex, source lines 3527–3550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07YE: sources/stacks-artin/pinned-original-source.tex, source lines 3709–3724. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CXT: sources/stacks-artin/pinned-original-source.tex, source lines 3257–3267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *This is Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. Assume*

(1) *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
(2) *$\mathcal{X}$ has (RS*),*
(3) *$\mathcal{X}$ is limit preserving,*
(4) *there exists an obstruction theory*[1]*,*
(5) *for an object x of $\mathcal{X}$ over $\mathrm{Spec}(A)$ and A-modules M_n, $n \geq 1$ we have*
   (a) *$T_x(\prod M_n) = \prod T_x(M_n)$,*
   (b) *$\mathcal{O}_x(\prod M_n) \to \prod \mathcal{O}_x(M_n)$ is injective.*

Then $\mathcal{X}$ satisfies openness of versality.

Proof. We prove this by verifying condition (4) of Lemma 0CXU. Let (ξ_n) and (R_n) be as in Remark 0CXT such that $\mathrm{Ker}(R_m \to R_n)$ is an ideal of square zero for all $m \geq n$. Set $A = R_1$ and $x = \xi_1$. Denote $M_n = \mathrm{Ker}(R_n \to R_1)$. Then M_n is an A-module. Set $R = \lim R_n$. Let

$$\tilde{R} = \{(r_1, r_2, r_3 \ldots) \in \prod R_n \text{ such that all have the same image in } A\}$$

Then $\tilde{R} \to A$ is surjective with kernel $M = \prod M_n$. There is a map $R \to \tilde{R}$ and a map $\tilde{R} \to A[M]$, $(r_1, r_2, r_3, \ldots) \mapsto (r_1, r_2 - r_1, r_3 - r_2, \ldots)$. Together these give a short exact sequence

$$(x, R \to A) \to (x, \tilde{R} \to A) \to (x, A[M])$$

of deformation situations, see Remark 07YE. The associated sequence of kernels $0 \to \lim M_n \to M \to M \to 0$ is the canonical sequence computing the limit of the system of modules (M_n).

Let $o_x(\tilde{R}) \in \mathcal{O}_x(M)$ be the obstruction element. Since we have the lifts ξ_n we see that $o_x(\tilde{R})$ maps to zero in $\mathcal{O}_x(M_n)$. By assumption (5)(b) we see that $o_x(\tilde{R}) = 0$. Choose a lift $\tilde{\xi}$ of x to $\mathrm{Spec}(\tilde{R})$. Let $\tilde{\xi}_n$ be the restriction of $\tilde{\xi}$ to $\mathrm{Spec}(R_n)$. There exists elements $t_n \in T_x(M_n)$ such that $t_n \cdot \tilde{\xi}_n = \xi_n$ by Lemma 07Y9 part (2)(b). By assumption (5)(a) we can find $t \in T_x(M)$ mapping to t_n in $T_x(M_n)$. After replacing $\tilde{\xi}$ by $t \cdot \tilde{\xi}$ we find that $\tilde{\xi}$ restricts to ξ_n over $\mathrm{Spec}(R_n)$ for all n. In particular, since ξ_{n+1} restricts to ξ_n over $\mathrm{Spec}(R_n)$, the restriction $\overline{\xi}$ of $\tilde{\xi}$ to $\mathrm{Spec}(A[M])$ has the property that it restricts to the trivial deformation over $\mathrm{Spec}(A[M_n])$ for all n. Hence by assumption (5)(a) we find that $\overline{\xi}$ is the trivial deformation of x. By axiom (RS*) applied to $R = \tilde{R} \times_{A[M]} A$ this implies that $\tilde{\xi}$ is the pullback of a deformation ξ of x over R. This finishes the proof. □

[1] Analyzing the proof the reader sees that in fact it suffices to check the functoriality (ii) of obstruction classes in Definition 07YG for maps $(y, B' \to B) \to (x, A' \to A)$ with $B = A$ and $y = x$.